

Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

Let's learn about data in CBIS!



Learning Outcome:

Understand information about relational database management system in spoken or written texts.

Ask your partner:

- Do you think data is always stored the same way? Why? Why not?
- How do you think computer-based data is different from other types of data?

Can Do Statements:

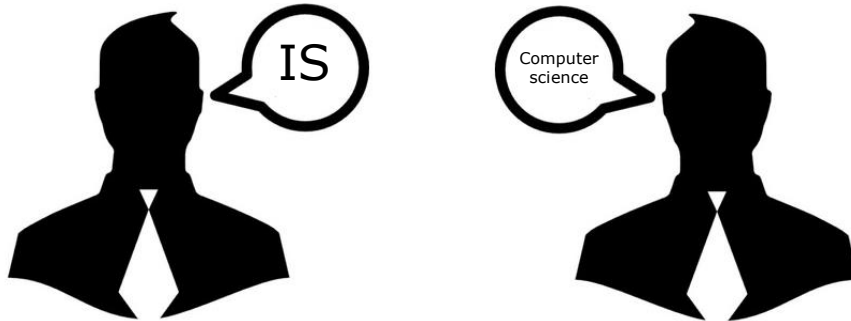
- I can give information about data in computer based information systems.
- I can talk about the requirements data must meet in a computer based information system.
- I can say how data resources are usually organized.





Let's Explore!

Don't you think we should get ready for the training session?



Instructions: Read the text below.

- Ryan:** Hey, Alice. The training on IS starts next Wednesday. Don't you think it is time to review some of the concepts to be ready?
- Alice:** Sure. But, why don't we talk to the manager? He always arranges a meeting with some of the experts to give us some guidelines.
- Ryan:** Don't you think we can get started instead of waiting for the manager to back us up? He's always complaining about how we cannot do things on our own.
- Alice:** You're right. Let's do it! We can't forget that we need to learn a lot about data in computer-based information systems ASAP so we can train ourselves the rest of the team.
- Ryan:** Exactly! That's why I'm worried. Let's start by getting some questions ready. The first one and most important is about computer-based information system. What is it?
- Alice:** That's an easy one. A computer-based information system is a type of system that uses computer technology to undergo some or all of its intended tasks. The simplest form of this system can be represented in a personal computer and software.
- Ryan:** Well answered. Now, are human resources included in a computer-based information system?
- Alice:** Yes, they are. On the one hand, we have the people in charge of making things work; for instance, IS specialists, system analysts, programmers and data administrators. On the other hand, we have the end users, which are the clients or users.
- Ryan:** What about data resources? Are they considered an important part of this system? Shouldn't we talk about it?
- Alice:** Of course, we should. Data resources include data and databases. Data is considered the raw material of information systems and can take many forms: the usual alphanumeric data, text data, image data, and audio data. Now, I have a question for you, Ryan. Are data resources required to meet any criteria?
- Ryan:** Yes, of course. They must be comprehensive, which means all the data regarding a particular subject is actually present in the database. There shouldn't be any redundancy, and data should be stored to minimize the cost of expected processing and storage as much as possible. One last question Alice, how are data resources of IS organized?
- Alice:** Well, they're organized into processed and organized data-databases and into knowledge in many forms, which can be facts, rules, and case examples related to successful business practices.
- Ryan:** Don't you think we're more than ready to give that training?
- Alice:** The manager is going to be so proud of us!



Let's Practice!

Main differences between computers and information systems

Instructions: Classify the sentences by writing the number of the sentence in the correct category.

Computer science and information systems have each one distinguishing feature that makes them different to each other. What are such characteristics?

1. They must be designed, maintained, and operated.
2. One of its concerns is the effective and efficient organization of data.
3. Requires understanding how the complete systems work, not just computers.
4. A system would not be complete without them.
5. Refers to the study of how computer hardware and software work together to transmit and process information.
6. They're one of the many tools of a more complex system.

Relevant to Computers and Computer Science	Relevant to Information Systems



Let's Practice!

Alice and Ryan prepare their questions about IS.

Instructions: Read the text and select all the simple present passive questions.

- Ryan:** Hey Alice, are we prepared to ask the IS manager for assistance regarding our questions for the training?
- Alice:** Well, not completely. Let's get some more questions ready before contacting him. What about this one: are data resources unprocessed content of information systems and databases?
- Ryan:** Sounds good. We can add: is data stored to reduce expenses? Is data meant to take many forms? Is data meant to include traditional alphanumeric information?
- Alice:** I guess those are enough for now. Don't you think so?
- Ryan:** Sure thing! Let's look for the IS manager now.





Extension Activity

I can know more about Data in Computer Based Information Systems.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more information about Data in Computer Based Information Systems.
- Create a post with the main characteristics of Data in Computer Based Information Systems.
- Share with your partners the post you created.

Self-Assessment

Yes No

- I look for more information about Data in Computer Based Information Systems. ☐ ☐
- I create a post with the main characteristics of Data in Computer Based Information Systems. ☐ ☐
- I share with your partners the post you created. ☐ ☐



Target Words

- **arrange** (verb) to organize, plan or prepare for something
e.g. He always arranges a meeting with some of the experts to give us some guidelines.
- **comprehensive** (adjective) with a large scope or content
e.g. They must be comprehensive, which means all the data regarding a particular subject is actually present in the database.
- **redundancy** (noun) a situation in which something is considered unnecessary and repeated
e.g. There shouldn't be any redundancy, and data should be stored to minimize the cost of expected processing and storage as much as possible.
- **storage** (noun) process of storing information in a computer memory
e.g. There shouldn't be any redundancy, and data should be stored to minimize the cost of expected processing and storage as much as possible.
- **alphanumeric** (adjective) codes that consist of alphabetical and numerical symbols
e.g. Data is considered the raw material of information systems and can take many forms: the usual alphanumeric data, text data, image data, and audio data.



Learning Points

Simple Present Passive Questions

The **passive voice** is used when the doer of an action is not necessary, is understood or is unknown. This voice is used in the **simple present** tense to talk about processes and actions that take place on a regular basis.

Now, we're going to learn how to ask **questions** using the **simple present** tense in the **passive** voice.

Remember that the **passive** form in any tense is always made of the verb to be and the past participle form of the main verb.

The verb **to be** usually goes in the tense we want to use; for our particular case this verb should go in its **simple present** tense form (*am, is, are*) and it is followed by the main verb in its **past participle** form. For the **interrogative** form, all you have to do is place the verb **to be** before the subject, as shown in the chart below:

Verb To Be	Subject	Past Participle	Complement + ?
Are	computers	designed	without considering information systems?

You can ask **wh- questions** when you require further information:

Wh- Question	Verb To Be	Subject	Past Participle	Complement + ?
Why	are	information systems	considered	more important than computers?

Examples

- *Are data resources raw material of information systems and database?*
- *Are data stored to minimize costs?*





Learning Points

Negative Questions to Express an Opinion

Negative Questions can be used as a conversation strategy.

Negative Questions is a helpful conversation strategy that will help you state opinions in a polite way, and also to suggest agreement with what you say.

Look at the following examples:

- **Don't** you think is cold today?
- **Shouldn't** we start planning our board meeting?
- **Haven't** you thought about checking this software out? It is really good.
- Why **don't** we start the meeting now, it's getting late?

These are not real questions, but they are asking the listener to agree with what they state.

Examples

- *Don't you think we can get started instead of waiting for the manager to get back to us?*
- *Why don't we ask about how data resources are usually organized? Isn't it a key aspect in IS?*



Learning Points

Difference Between Computers and Information Systems

The fields of **computer science** and **information systems** have some differences; the section below explains their main characteristics.

Computers are effective and efficient ways to process data, and engineers and software developers consider them a necessary part of an **information system**. However it is also said that an **Information System (IS)**, involves much more than just **computers**

Information System (IS)	Computers
Requires an understanding of the business and its environment. </br> It is necessary to understand the procedures related to buying and selling stocks, bonds, options, among others, including irregular demands made on the system, as well as all government-related regulations.	Computers are only one part of a complex system that must be designed, operated, and maintained. </br> Computers, are one component of a much bigger and more complex system. </br> Computers act as a complement for the whole system, without computers the system would be incomplete.

Examples

- We could ask him to discuss the difference between computers and information systems why don't you include this as part of the meeting, IS requires an understanding of the business and its environment, while Computers are only one component in a complex system.*

